Explicit semidefinite representations and local certificates for Q222

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We use the eight coordinates

$$q = (a_0, a_1, b_0, b_1, E_{00}, E_{01}, E_{10}, E_{11}).$$

The recovered evaluated matrices give an exact representation of a two-dimensional projection, two explicit local feature constructions, and four full-coordinate outer approximations. They do not give an evaluated exact lift of the full eight-coordinate set. This distinction is independent of the qualitative finite-lift existence theorem.

1 An exact two-coordinate projection

Put $t = a_0$ and $s = E_{00} + E_{01} + E_{10} - E_{11}$.

Theorem 1. *The image of $\mathcal{Q}_{222}$ in these coordinates is*

$$\mathcal{T} = \{(t, s) : |t| \le 1, \quad t^2 + s^2/4 \le 2\}.$$

It has the explicit order-four representation

$$D(t, s) = \operatorname{diag}\left(\begin{pmatrix} \sqrt{2}+t & s/2 \\ s/2 & \sqrt{2}-t \end{pmatrix}, 1-t, 1+t\right) \succeq 0. \tag{1}$$

There are no auxiliary variables. All coefficients belong to $\mathbb{Q}(\sqrt{2})$, with $\sqrt{2}$ the positive root of $x^2 - 2$.

Proof. Binary observables can be dilated to reflections with the same local embedding, preserving all eight expectations. For reflections define

$$\begin{aligned} S &= A_0(B_0 + B_1) + A_1(B_0 - B_1), \\ S' &= A_0(B_0 - B_1) + A_1(B_0 + B_1), \\ T_\alpha &= \alpha A_0 + S, \qquad R_\alpha = \alpha A_1 - S'. \end{aligned}$$

Reflection relations and cross-party commutation give

$$T_\alpha^2 + R_\alpha^2 = (8 + 2\alpha^2)\mathbb{I}.$$

Indeed $S^2 + (S')^2 = 8\mathbb{I}$, and the linear terms cancel since $\{A_0, S\} = \{A_1, S'\}$. With $\beta = \sqrt{8 + 2\alpha^2}$ this gives

$$\beta\mathbb{I} - T_\alpha = \frac{(\beta\mathbb{I} - T_\alpha)^2 + R_\alpha^2}{2\beta} \succeq 0.$$

Thus $\alpha t + s \le \sqrt{8 + 2\alpha^2}$. For $|t| \le 1$, take $h = \sqrt{2 - t^2}$ and $\alpha = 2t/h$. The result is $s \le 2h$. Flipping both Bob outcomes gives $-s \le 2h$.

Conversely fix $t \in [-1,1]$, put $r = \sqrt{1-t^2}$, and take

$$|\psi_t\rangle = \sqrt{\frac{1+t}{2}}|00\rangle + \sqrt{\frac{1-t}{2}}|11\rangle, \quad A_0 = Z, \quad A_1 = X, \quad B_{0,1} = \frac{Z \pm rX}{h}.$$

These are reflections, and their behavior is

$$\left(t, 0, \frac{t}{h}, \frac{t}{h}, \frac{1}{h}, \frac{1}{h}, \frac{1-t^2}{h}, -\frac{1-t^2}{h}\right).$$

It projects to $(t, 2h)$. The Bob flip realizes $(t, -2h)$; a finite classical mixture realizes every intermediate s. At $t = \pm 1$ the same construction remains defined since $h = 1$.

Finally (1) forces $|t| \le 1$. On this interval its two-by-two diagonal entries are strictly positive, and its determinant is $2 - t^2 - s^2/4$. These conditions prove both matrix inclusions. □

In the original eight coordinates the coefficient matrices of (1) are particularly simple. Write e_{ij} for matrix units with indices $1, \dots, 4$. Then

$$A_0 = \operatorname{diag}(\sqrt{2}, \sqrt{2}, 1, 1), \quad A_{a_0} = \operatorname{diag}(1, -1, -1, 1),$$
$$A_{E_{00}} = A_{E_{01}} = A_{E_{10}} = \tfrac{1}{2}(\mathsf{e}_{12} + \mathsf{e}_{21}), \quad A_{E_{11}} = -\tfrac{1}{2}(\mathsf{e}_{12} + \mathsf{e}_{21}),$$

and $A_{a_1} = A_{b_0} = A_{b_1} = 0$. Their feasible set in $\mathbb{R}^8$ is the inverse image of $\mathcal{T}$, not $\mathcal{Q}_{222}$.

1.1 Fixed features and an auxiliary-variable version

On $\mathcal{T}$ define the five fixed testing functions

$$u_1 = \sqrt{\sqrt{2} + t}, \quad u_2 = s/(2u_1), \quad u_3 = \sqrt{\sqrt{2} - t - u_2^2}, \quad u_4 = \sqrt{1-t}, \quad u_5 = \sqrt{1+t}.$$

Their roots are defined on the whole set because $\sqrt{2} + t \ge \sqrt{2} - 1 > 0$ and $(\sqrt{2} + t)u_3^2 = 2 - t^2 - s^2/4 \ge 0$. The within-block products in the spaces (u_1, u_2), (u_3), (u_4), (u_5) are exactly

$$u_1^2 = \sqrt{2} + t, \quad u_1 u_2 = s/2, \quad u_2^2 = y, \quad u_3^2 = \sqrt{2} - t - y, \quad u_4^2 = 1 - t, \quad u_5^2 = 1 + t.$$

Thus an extracted lift with $N = 5, m = 1$ is

$$L(t, s, y) = \operatorname{diag}\left(\begin{pmatrix} \sqrt{2} + t & s/2 \\ s/2 & y \end{pmatrix}, \sqrt{2} - t - y, 1 - t, 1 + t\right). \tag{2}$$

For $(t, s) \in \mathcal{T}$, the witness $y = s^2/[4(\sqrt{2}+t)]$ makes every block positive semidefinite. Conversely (2) gives $s^2/[4(\sqrt{2} + t)] \le y \le \sqrt{2} - t$ and $|t| \le 1$, which imply the defining ellipse inequality. This also proves the elimination of y without a closure argument.

The same fixed functions certify every affine support. For a source tilt α and sign $\epsilon = \pm 1$, put

$$\gamma = \sqrt{4 + \alpha^2}, \quad v_0 = \sqrt{(\gamma - \alpha)/2}, \quad v_1 = -\epsilon\sqrt{(\gamma + \alpha)/2}.$$

Then $v_0 v_1 = -\epsilon$ and, in separately labelled real closed source and testing fields,

$$\sqrt{2}\gamma \otimes 1 - \alpha \otimes t - 1 \otimes \epsilon s = (v_0 \otimes u_1 + v_1 \otimes u_2)^2 + (v_1 \otimes u_3)^2.$$

The support value is $h(\alpha) = \sqrt{8 + 2\alpha^2}$ for $|\alpha| \le 2$, and $h(\alpha) = 2 + |\alpha|$ otherwise. For the latter range, $\sigma = \operatorname{sgn}\alpha$ gives

$$2 + |\alpha| - \alpha t - \epsilon s = 4 - 2\sigma t - \epsilon s + (|\alpha| - 2)(1 - \sigma t).$$

The first term uses the displayed square identity at $\alpha = 2\sigma$; the last uses u_4^2 or u_5^2. Positive constants use $1 = (u_4^2 + u_5^2)/2$. Scaling these slacks and adding a positive constant covers every affine function nonnegative on $\mathcal{T}$. This includes unbounded source tilts and all tilt endpoints without identifying the two field copies.

1.2 An exact obstruction to treating this as the full set

The rational behavior

$$q_* = (0,0,0,0,1,1,1/4,-1/4)$$

has nonnegative nonsignalling probabilities $p(a,b|x,y) = (1+abE_{xy})/4$. Its maximum absolute CHSH value over the eight sign patterns is $5/2 < 2\sqrt{2}$. Every marginal is zero, so it satisfies every standard relabelled single-marginal tilted-CHSH inequality, whose bound is at least $2\sqrt{2}$. It also lies in (1), since $t = 0, s = 5/2$ and $s^2/4 = 25/16 < 2$.

It is not quantum. After dilation and purification, $E_{00} = E_{01} = 1$ would imply

$$\|(A_0 - B_y)\psi\|^2 = 2 - 2E_{0y} = 0 \quad (y = 0,1).$$

Hence $(B_0 - B_1)\psi = 0$, forcing $E_{10} - E_{11} = 0$, whereas the displayed point gives $1/2$. Thus intersecting all these projected lifts and the probability inequalities still does not recover $\mathcal{Q}_{222}$.

2 An evaluated regular-C2 local certificate

The following construction recovers a complete local coefficient lift. Its domain is part of the assertion. For the seven-source order (d,n,m,C,A,D,α), the full quarter-parent is

$$\begin{aligned}\widetilde{P} = {} & nD_uD_vX^2 + m(vW - X + uvY - uZ)^2 + CD_v(uW - Y)^2 + AD_uD_vY^2 \\ & + (D + d/2)(uW + uvX - Y - vZ)^2 + (\alpha + d/2)D_u(vY - Z)^2 \\ & + (d/2)D_v(X + uZ)^2 - (d/2)D_uD_vZ^2, \qquad D_u = 1 + u^2, \quad D_v = 1 + v^2.\end{aligned}$$

Set only $d = Z = 1$, and write $P_c(z)$, where $c = (n,m,C,A,D,\alpha)$ and $z = (u,v,W,X,Y)$. The exact anchors are

$$c_* = (7/8, 9/8, 1/18, 11/8, 9/8, 1/18), \qquad \theta_* = (1,1,1,-1/10,1/10).$$

Let $F(c,\theta) = (P_c, \partial_u P_c, \partial_v P_c, \partial_W P_c, \partial_X P_c, \partial_Y P_c)(\theta)$. Put $s = (c - c_*, h)$, $h = \theta - \theta_*$, and independently $t = z - \theta_*$. The source domain and testing box are exactly

$$F(c,\theta) = 0, \quad |s_i| \le \rho \ (i = 1,\dots,11), \qquad |t_j| \le \rho \ (j = 1,\dots,5), \qquad \rho = \frac{510101005}{350806159660744}.$$

The value equation supplies the remaining homogeneous Z derivative by Euler's identity. The six-by-six source Jacobian has determinant $114048/78125$ at the anchor. On this entire contact box its relative row-norm perturbation is at most $11650604933999/11225797109143808 < 1$, so the source solution is unique for any fixed contact in the box. No global cover is implied.

Here is an exact finite prescription for all the features and matrices. Expand $P_c(\theta + \Delta)$ and remove degree zero and one in Δ. Allocate each remaining monomial to a symmetric matrix H: use the first coordinate with exponent at least two as a diagonal square; if none exists, use the first two occupied coordinates and put half the coefficient in each symmetric off-diagonal entry. Then

$$P_c(\theta + \Delta) = P_c(\theta) + \nabla P_c(\theta)^T\Delta + \Delta^T H \Delta$$

is an exact polynomial identity with no Taylor remainder. At the anchor $H_0 = LD_0L^T$, where

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ -9801/10201 & 1 & 0 & 0 & 0 \\ 1210/10201 & 8991/20002 & 1 & 0 & 0 \\ 10080/10201 & -5040/10001 & 79/1010 & 1 & 0 \\ -10120/10201 & 5060/10001 & -119/1010 & 2/81 & 1 \end{pmatrix},$$

$$D_0 = \operatorname{diag}\left(\frac{10201}{3600}, \frac{20002}{91809}, \frac{27775}{10001}, \frac{891}{202}, \frac{1067}{162}\right).$$

Thus $f = L^T z$, $\widehat{H} = L^{-1} H L^{-T}$ give the correctly oriented identity $\Delta^T H \Delta = (\Delta f)^T \widehat{H} \Delta f$. Put $\varepsilon = 10001/918090$. Use the 25 linear forms $\ell_k = f_i$ and $f_i \pm f_j$, in that order, with coefficients

$$\widehat{H}_{ii} - 8\varepsilon, \qquad \varepsilon + \widehat{H}_{ij}/2, \qquad \varepsilon - \widehat{H}_{ij}/2.$$

Each coordinate participates in four pairs, so the pair diagonal contribution is exactly 8ε. Substitute $\Delta = t - h$ and write the resulting coefficient polynomials as $\gamma_k(s,t)$. Their constants are at least ε, and their largest nonconstant coefficient norm is exactly

$$B_* = \frac{43803942777143}{46827180450}, \qquad \rho = \frac{\varepsilon}{8(1+B_*)} < \tfrac{1}{2}.$$

All these rational coefficient data are given in the accompanying `regular-c2-local/regular_c2_frames.json`.

For completeness, the separation of every coefficient into nonnegative pure factors has a uniform exact formula. Group its support uniquely as

$$\gamma = \gamma_0 + G(s) + T(t) + \sum_{\beta \neq 0} A_\beta(s) t^\beta,$$

where G, T, A_β have zero constant. Define

$$\sigma_\beta = \sum_a |[s^a] A_\beta|, \quad U_\beta = A_\beta + \sigma_\beta \rho, \quad V_\beta = t^\beta + \rho^{|\beta|}, \quad B = \sum_\beta \sigma_\beta \rho^{|\beta|+1},$$

$$p(s) = \gamma_0/2 - B/2 + G(s) - \sum_\beta \rho^{|\beta|} A_\beta(s),$$
$$q(t) = \gamma_0/2 - B/2 + T(t) - \sum_\beta \sigma_\beta \rho\, t^\beta.$$

Expansion proves $\gamma = p + q + \sum U_\beta V_\beta$. On the independent boxes, $U_\beta, V_\beta \geq 0$. If g, r, M are the coefficient norms of $G, T, \sum A_\beta t^\beta$, then $g + r + M \leq B_*$, $B \leq \rho M$, and

$$p, q \geq \gamma_0/2 - \tfrac{3}{2}\rho(g + r + M) \geq 5\varepsilon/16 > 0.$$

These inequalities prove all root signs, including box boundaries.

For each frame and each pair $(U, V) = (p, 1), (1, q), (U_\beta, V_\beta)$, use the testing vector and source matrix

$$\psi_b(t) = \sqrt{V(t)} \begin{pmatrix} 1 \\ \ell_k(t) \end{pmatrix}, \qquad Q_b(s) = U(s) \begin{pmatrix} \ell_k(h)^2 & -\ell_k(h) \\ -\ell_k(h) & 1 \end{pmatrix}.$$

Then $Q_b \succeq 0$ and $\psi_b^T Q_b \psi_b = UV(\ell_k(t) - \ell_k(h))^2$. The 735 mixed pairs and 50 pure pairs give exactly 785 blocks of order two and 1570 listed testing functions. Contact makes the discarded constant and linear Taylor terms zero, proving the full-parent identity. The same proof is a tensor sum of squares in independently labelled real closed fields because each square root is taken in its own field.

The within-block products are $V, V\ell_k, V\ell_k^2$. Let $m_0, \ldots, m_{236}$ be the lexicographically ordered union of their monomial supports. Define exact rational arrays by coefficient extraction

$$\psi_{b,i}\psi_{b,j} = \sum_k p_{b,ij,k} m_k, \qquad \widetilde{P}_\lambda(\theta_* + t, 1) = \sum_{k,\nu} T_{k\nu} \lambda_\nu m_k, \quad \lambda = (d, n, m, C, A, D, \alpha).$$

The evaluated data contain 27747 nonzero product coefficients and 295 nonzero target coefficients. The exact coefficient lift is

$$\sum_{b,i\le j}(2-\delta_{ij})p_{b,ij,k}G_{b,ij}=\sum_{\nu}T_{k\nu}\lambda_\nu\quad(k=0,\dots,236),\qquad G_b\succeq 0.$$

It has 2355 upper-triangular Gram variables and 237 equations. Every feasible point gives the displayed nonnegative polynomial identity; every source in the contact box has the explicit feasible matrices above. Equality with the entire cone of polynomials nonnegative on the box is not asserted. This is a local certificate lift in source coordinates, not an eight-coordinate quantum-set lift.

3 A second recovered local feature library

The September C1D construction gives a different explicit finite local library. Its source domain is

$$0<a\le 1,\quad 0<b,q\le\epsilon,\quad 0<r<1,\quad 0<E\le 1,\qquad \epsilon=1/2788018,$$

with independent tests $|u|,|v|,|W|,|X|,|Y|\le\epsilon$, $Z=1$. Put $e^2=E$, $r_r^2=r$, $s_s^2=1-r$ with positive roots. Its five separated roots are

$$\begin{pmatrix} uW-b^2v\\ buv\\ b^2(v^2-q^2)\\ (b/e)(X+u)-a^2b^3q^2e\,vW\\ (Y-ruW)/(r_rs_s)\end{pmatrix}.$$

The recovered full source Gram is $8I_5$ plus a polynomial correction whose upper coefficient norm is 1394008. Each correction monomial contains a small source or testing variable. Hence its row perturbation is less than $1/2$ on this rectangle, uniformly without lower bounds on $E,r,1-r$.

For a scalar channel $\rho+\sum\beta_mC_m\otimes T_m$, set $S=\sum|\beta_m|$. Choose the small pure factor as t_m, absorb the sign into it, and call the other factor l_m. Each term has $|t_m|\le\epsilon$, $|l_m|\le 1$, and $\rho/S>\epsilon$. The exact identity

$$\rho/S+t_m\otimes l_m=\tfrac12\{(\rho/S+t_m)\otimes(1+l_m)+(\rho/S-t_m)\otimes(1-l_m)\}$$

uses nonnegative factors on their own sides. The 25 diagonal and pair channels produce 82928 tensor squares. The data specify all 41464 term pairs and all pure-factor radical choices.

After adjoining the 1212 testing radicals, the fixed testing spanning functions are the 3844 listed pairs of

$$\kappa\zeta_j\chi_a,\qquad \chi_a\in\{1,uW,v,uv,v^2,X+u,vW,Y\},$$

where $\kappa^2(1+u^2)(1+v^2)(1+W^2+X^2+Y^2)=1$. Box equations $\mathrm{box}_t^2+t^2=\epsilon^2$ and the printed radical equations give 1223 algebra variables and 1218 equations. The testing image is this entire closed local box. The list is not claimed linearly independent. Neither its union with the regular-C2 library nor its own local identities establishes a finite global cover. The original polynomial allocation multipliers and pure-factor restoration are preserved beside the library; they are required to identify its source Gram with the original Bell slack.

4 Evaluated full-coordinate outer pencils

The recovered data include the following actual rational matrices. Here N is the total block-diagonal order and m is the auxiliary dimension; none of these rows is an evaluated exact full-set lift.

N	m	nonzero records	reverse inclusion
8368	16480	55712	disproved
10736	21472	72608	disproved
13744	29920	153248	unproved
63888	31111	455750	unproved (C160)

For the first three files a record $[k, i, j, a]$ adds a to the upper entry (i, j) of A_0 if $k = 0$, of A_k if $1 \le k \le 8$, and of B_{k-9} otherwise. Mirror each off-diagonal once. The represented set is

$$\mathcal{O}_N = \{q : \exists y \in \mathbb{R}^m, A_0 + \sum_{k=1}^{8} q_k A_k + \sum_{j=0}^{m-1} y_j B_j \succeq 0\}.$$

Here the auxiliary variables and matrices are indexed from zero, so a coefficient index $k = 9 + j$ multiplies $y_j B_j$. C160 instead stores $[b, i, j, k, p, q]$, meaning the entry p/q in block b and coefficient k with the same visible/auxiliary convention. Its block orders and all 455750 records are supplied explicitly.

The first three constructions dualize finite cones of globally valid source certificates, with exact source maps, all source signs, and explicitly positive strict source points. An elementary adjoint argument then proves $\mathcal{Q}_{222} \subseteq \mathcal{O}_N$: if $L(c, z)$ is a source certificate pencil with a strict positive point, the image $\{L^*(Y) : Y \succeq 0\}$ is closed. Indeed bounded adjoint images, paired with the strict source point, bound $\operatorname{tr} Y$; subsequential limits give closedness. Bipolarity identifies the dual of the projected source cone with the attained adjoint image. Exact linear elimination supplies every affine auxiliary coordinate in the recovered primal arrays. Thus source-certificate soundness gives the quantum forward inclusion, not its converse.

For $N = 13744$ the source certificates use four blocks of orders $(14, 15, 14, 15)$ for C1 and $(15, 14, 14, 15)$ for C2 in the full identity

$$(1 + u)(1 + v)F_c = \sum_{g \in \{0,1\}^2} u^{g_1} v^{g_2} f_g^T G_g f_g, \quad G_g \succeq 0.$$

The multiplier is pure testing and positive for $u, v > 0$; state sign gauges and homogeneous limits give the other charts. The global pencil retains 128 C1 copies, 64 C2 copies, 64 adjacent and 32 opposite exact closed-face copies, and 16 event inequalities. Their local orders and auxiliaries give exactly $N = 13744, m = 29920$, with 2928 blocks of maximum order 15. Adequacy of these fixed features for every physical source is not established by their coefficient matching identities.

4.1 Why the two smaller global candidates are strictly outer

The exact physical C1 source is

$$d = 1, \quad m = \frac{1166523000}{31661820415017653}, \quad C = \frac{322803000}{186199784281},$$
$$A = \frac{218288946200}{1359122513}, \quad D = \frac{90943434293}{222507771414}, \quad \alpha = \frac{248802430236}{3300726103},$$
$$\beta = \frac{205476973}{51646655494}.$$

It has the contact $(u, v; \psi) = (1/13, 16; 140, -1/32, 1/50, 1)$. Its complete state-kernel and angular equations hold exactly. Its six positive Laurent determinant weights balance around zero, so the elementary weighted AM–GM identity proves determinant nonnegativity. The positive m, C, A rows have a strict complementary state block; the Schur complement therefore proves physicality, including signs and homogeneous endpoints.

Let $f_* = J_1 c_*$, where

$$\begin{aligned}(J_1c)_0 &= d + m + C + A + D + \alpha + \beta,\\ (J_1c)_{1:4} &= (-A - \alpha, m - C - D + \beta, C + A - \beta, -m + D - \alpha),\\ (J_1c)_{5:8} &= (-d/2 - A, d/2 + \alpha, -d/2 - C - \beta, -d/2 - m - D).\end{aligned}$$

The four kernel equations, two angular equations and two deterministic zeros have rank eight in the nine affine coefficients: the indicated eight-column minor is

$$\frac{4090727687944001770960311645728 64}{10770194675703125} \neq 0.$$

In any positive physical decomposition every summand obeys these same eight equations. It is therefore proportional to f_*, with a nonnegative scalar. Thus f_* is an extreme physical ray.

For the actual order-8368 and order-10736 source cones the only nonnegative component preimages of this ray are c_* and $\sigma_1 c_* = (d, C, m, \alpha, \beta, A, D)$. The complete source-map arrays retain the finite screen establishing this statement. The preserved positive rational dual matrices give, on the actual C1 coefficient pencil, the trace identities

$$\sum_b \operatorname{tr}(Y_b^{(0)} L_b(c, y)) = \ell(c) - d, \qquad \sum_b \operatorname{tr}(Y_b^{(1)} L_b(c, y)) = \ell(\sigma_1 c) - d,$$

where $\ell(c) = 93424\, \psi_*^T P_c(u_*, v_*)\psi_* - d$ and $\ell(c_*) = -1$. Every auxiliary coefficient is zero, and all dual matrices are positive definite. The Gram trace at the proposed source is therefore -2: the further subtraction of d is the actual scalar certificate coefficient, not a change of parent normalization. Hence neither permitted preimage lies in its certificate cone.

These finite source sums are closed: positive finite evaluation moments bound all Gram matrices of convergent certificate sources, and the positive affine constant bounds each nonnegative component in the sum. Thus separating f_* from the closed source sum gives a point of its dual violating f_*. The event rays force its constant coordinate to be positive, so it normalizes to $(1, q)$ with $q \notin \mathcal{Q}_{222}$. Consequently

$$\mathcal{Q}_{222} \subsetneq \mathcal{O}_{8368}, \qquad \mathcal{Q}_{222} \subsetneq \mathcal{O}_{10736}.$$

This proof gives strictness; it does not print one numerical behavior and thousands of auxiliary coordinates. It does not refute the enlarged order-13744 or the separate C160 candidate.

4.2 The C160 moment construction and its coverage dependency

For the fourth candidate use angular parameters $z = (k, x_0, y_0, x_1, y_1) \in [0, 1]^5$ with

$$ky_0(1 - x_0) \geq x_0(1 - y_0), \quad kx_1(1 - y_0) \geq y_0(1 - x_1), \quad ky_1(1 - x_1) \geq x_1(1 - y_1).$$

Call this domain T. Put $d(t) = (1 - t)^2 + t^2$, $c(t) = (1 - 2t)/d(t)$, $s(t) = 2t(1 - t)/d(t)$ and $D = (1 + k^2)d(x_0)d(y_0)d(x_1)d(y_1)$, so $1/16 \leq D \leq 2$. The physical map is

$$\Phi_{A_x} = \frac{1 - k^2}{1 + k^2} c(x_x), \quad \Phi_{B_y} = \frac{1 - k^2}{1 + k^2} c(y_y), \quad \Phi_{E_{xy}} = c(x_x)c(y_y) + \frac{2k}{1 + k^2} s(x_x)s(y_y).$$

Its stated full-quantum coverage is

$$\mathcal{Q}_{222} = \operatorname{conv}\big(\{\ell_1, \ldots, \ell_{16}\} \cup \bigcup_{g \in \mathcal{G}} g\Phi(T)\big), \tag{3}$$

where $\mathcal{G}$ contains the 128 local relabellings and party swaps. The recovered construction invokes the Barizien–Bancal extremal characterization [1] for (3). This external coverage theorem

and its coordinate interface are a dependency of its full-$\mathcal{Q}_{222}$ forward inclusion; they were not independently re-proved in this extraction. The following moment statement itself is direct.

Let $p_e = \prod_i p_{e_i}(z_i)$ for $(p_0, p_1, p_2) = ((1-t)^2, t(1-t), t^2)$ and $e \in \{0,1,2\}^5$. There are 243 moments. The 160 printed feature blocks have entries $\sqrt{w_b/D}$ times products of z_i or $1 - z_i$, where every w_b is a nonnegative rational multiple of the cube factors and the three displayed domain inequalities. Their orders sum to 499. The source cache prints all product coefficients in the p_e basis and all nine target columns $D, -D\Phi_1, \ldots, -D\Phi_8$. For $z \in T$, the exact witness $Y_e = p_e(z)/D(z)$ makes every moment block the rank-one Gram of its feature vector, and the target map gives $(1, \Phi(z))$. Nine independent target equations eliminate nine moments, leaving 234 canonical auxiliaries. Thus $\operatorname{conv} \Phi(T)$ lies in the canonical order-499 projection.

Block zero has weight one, includes all five variables, and its products span all 243 moments. The exact constant-column Gram

$$Q_D = \begin{pmatrix} I_{16} & I_{16} \\ I_{16} & 2I_{16} \end{pmatrix} \succ 0$$

bounds the trace of its PSD moment block at mass one and forces every moment to vanish at mass zero. Therefore the canonical projection is compact and its homogenization introduces no nonzero zero-mass point. Taking 128 copies and 16 nonnegative deterministic weights, then eliminating nine independent weights, gives $N = 128 \cdot 499 + 16 = 63888$ and $m = 128 \cdot 243 + 7 = 31111$. This proves the forward inclusion subject to (3). The reverse inclusion would require a universal adequacy theorem for this particular fixed 160-block space; neither the Dirac witnesses nor exact elimination supplies it.

5 Size bounds and exact algebraic search

The full quantum set has a rational first-order membership formula $\mathrm{Phys}(q)$ using nine real two-qubit atoms: each atom has a unit real four-state vector and two unit measurement circles, and its eight outputs are Pauli expectation polynomials. Nine convex weights suffice by Carathéodory in $\mathbb{R}^8$. Realification and direct sums cover complex mixed models. The compact atom domain and weight simplex give compactness of $\mathcal{Q}_{222}$. The formula uses 81 existential variables, 36 equations, nine inequalities and degree at most five.

We first exclude small pencils and remove redundant auxiliary variables. The order below is that of a single PSD cone; a product of blocks embeds into a block-diagonal cone whose order is the sum of the block orders.

Proposition 2. *Every affine semidefinite lift of $\mathcal{Q}_{222}$ through Sym_N^+ has $N \geq 9$. A pencil of that order needs at most $D - 9$ auxiliary directions, where $D = N(N+1)/2$.*

Proof. For output bits $a, b \in \{0,1\}$, the sixteen affine functionals

$$p(a,b|x,y) = \tfrac{1}{4}\big(1 + (-1)^a a_x + (-1)^b b_y + (-1)^{a+b} E_{xy}\big)$$

are nonnegative on $\mathcal{Q}_{222}$. Write a deterministic assignment as the four bits (r_0, r_1, s_0, s_1); its event probability is one exactly when $r_x = a$ and $s_y = b$. Starting with $F_0 = \mathcal{Q}_{222}$, successively impose the zero probabilities in the table. Each witness obeys all preceding zero conditions and has the newly listed probability equal to one.

k	added condition	removed witness	remaining deterministic vertices
1	$p(1,0\|0,0)=0$	1000	12
2	$p(0,1\|0,0)=0$	0010	8
3	$p(1,1\|0,1)=0$	1011	6
4	$p(1,1\|1,0)=0$	1110	5
5	$p(1,1\|0,0)=0$	1010	4
6	$p(1,1\|1,1)=0$	0101	3
7	$p(0,1\|0,1)=0$	0001	2
8	$p(1,0\|1,0)=0$	0100	1

Each F_k is an exposed face, being the zero set of a sum of nonnegative probabilities, and contains the deterministic assignment 0000. Thus $F_0 \supsetneq F_1 \supsetneq \cdots \supsetneq F_8$ is a chain of nine nonempty faces. The constant and the eight coordinate functions on the sixteen assignments are distinct orthogonal sign characters, so $\mathcal{Q}_{222}$ has affine dimension eight. Hence a nonempty face chain cannot have more terms. The exact Boolean checker enumerates all 2^{16} event subsets and confirms this maximal chain.

Now let $\mathcal{Q}_{222} = \pi(T)$, where $T = \mathcal{A} \cap \mathrm{Sym}_N^+$ for an affine subspace $\mathcal{A}$ and an affine map π. For a nonempty face C of T put $U(C) = \sum_{X \in C} \operatorname{range} X$. An average of finitely many points of C is positive definite on $U(C)$, because the kernel of a sum of PSD matrices is the intersection of their kernels. If nonempty faces $C \subsetneq C'$ had the same support, choose such an $X \in C$. For any $Y \in C'$ and sufficiently small $\theta \in (0,1)$,

$$Z = \frac{X - \theta Y}{1-\theta} \in T, \qquad X = \theta Y + (1-\theta) Z.$$

The face property would force $Y \in C$, a contradiction. Therefore proper face inclusions strictly increase the support dimension. This argument uses only the actual support and does not require a positive definite point in the ambient cone.

Moreover $0 \notin T$: otherwise $\mathcal{A}$ would be linear and T a cone, so its affine image could be compact only if it were a singleton. The inverse images $T_k = \{X \in T : \pi(X) \in F_k\}$ are strictly nested nonempty faces. Their support dimensions consequently satisfy

$$N \geq \dim U(T_0) > \dim U(T_1) > \cdots > \dim U(T_8) \geq 1,$$

which proves $N \geq 9$.

Finally write a pencil as $A_0 + \sum_{i=1}^{8} q_i A_i + \sum_j y_j B_j$ and replace the B_j by a basis of their span V, of dimension r. Compactness makes $A_1, \ldots, A_8$ independent modulo V: any nonzero relation would give a full feasible line in q while keeping the matrix fixed. Also $A_0 \notin \operatorname{span}(A_1, \ldots, A_8, V)$. Otherwise the matrix would vanish at some (q_*, y_*), and scaling any feasible matrix by $t \geq 0$ would give $q_* + t(q - q_*) \in \mathcal{Q}_{222}$. Compactness would make $\mathcal{Q}_{222}$ a singleton. Thus $r + 9 \leq D$, proving the auxiliary bound. □

Fix $N \geq 9$ and put $p_N = D - 9$. Let c contain the entries of the $9 + p_N$ symmetric coefficient matrices. There are $(9 + p_N)D = D^2$ unknown coefficients. Quantifier elimination decides the rational sentence

$$\exists c \in \mathbb{R}^{D^2} \ \forall q \in \mathbb{R}^8 \quad [\mathrm{Phys}(q) \iff \exists y \in \mathbb{R}^{p_N} : L_c(q,y) \succeq 0], \tag{4}$$

with PSD replaced by nonnegativity of all principal minors. A positive decision permits exact real algebraic coefficient witnesses with real-root selection data; transfer back to $\mathbb{R}$ gives the same identity for all real q. Given the qualitative finite-lift theorem, enumeration over $N = 9, 10, \ldots$ terminates. The first unexcluded template has $N = 9$, $p_N = 36$ and 2025 unknown coefficients. Its feasibility is not asserted, and no numerical upper order or coefficient tuple has yet been returned by this search.

The stored three-block order-two input has total order six, nine auxiliaries and 162 unknown coefficients. Proposition 2 proves its sentence false: it is excluded, although its historical filenames retain `UNSOLVED`. The lower bound concerns total PSD order and leaves the possibility of more blocks of order two undecided.

The tensor criterion supplies existence of a finite global testing space without an evaluated dimension here. Completing an explicit global construction requires selected testing functions, their product basis and dimensions (E1), one fixed original-coordinate pencil (E2), and both full-set inclusions for that pencil (E3). The algebraic search is an alternative route to E2–E3 once it returns a coefficient tuple.

Scientific data and exact checks

The accompanying `extracted-lifts/` directory contains the exact projection matrices and checker; the complete regular-C2 feature and product arrays with an independent standard-library checker; the C1D local square library, source multipliers and algebra; and all four global outer matrices with source maps and separator data. Its README specifies the two COO conventions and the status of each construction. In the two disproved candidate files only scientific scope metadata was updated; all coefficient and block arrays are unchanged. Provenance records bind the source bytes and the copied mathematical data. No matrix here is presented as an exact full-$\mathcal{Q}_{222}$ lift.

References

[1] V. Barizien and J.-D. Bancal, *Quantum statistics in the minimal scenario*, arXiv:2406.09350 (2024), Theorem 1 and its full-alternance parametrization.